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Project Overview

In this project, I aim to be the first to measure the individual-level effect of winning the H-1B lottery on labor market outcomes by linking administrative data on H-1B lottery results to individual-level outcomes in LinkedIn. The empirical challenge lies in the linkage: I have very limited information on individuals in the H-1B lottery data (firm, year of birth, gender, country of nationality). Past work has been impeded by the idea that if an individual loses the H-1B lottery, they likely don't end up working at the firm listed on the lottery application and are therefore not linkable via firm. My contribution is to show that for the subset of applicants that graduated with a STEM degree in the U.S., the three-year OPT program permits working at a firm and continuously applying for the H-1B lottery until three years has expired. This allows me to link on firm for OPT-eligible individuals.

Summary of Progress so far

Over the summer, my goal was to try to finish linking (however imperfect) to try to get some initial results and decide on the direction of the project. Linking proved much more difficult than initially expected and my final linked sample as of now exhibits two primary problems: selection/external validity and noise (applications are being matched to multiple potential LinkedIn users).

With these caveats in mind, my empirical specification for H-1B application j with company c in year t is as follows:

$$y_{jct} = \beta_0 + \beta_1 win_j + \beta_2 ADE_j + \varepsilon_{jct}$$

where win_j represents the outcome of lottery application j for an individual at company c in lottery year t, ADE_j is an indicator for advanced degree status, which increases lottery odds, and y_{jct} is the outcome of interest. In Table 1 (Table 2) I present indicators for working at the same company, working in the US, and returning to school one (two) year(s) after the lottery. Odd columns follow the above specification, even columns include company by year fixed effects. The estimates largely suggest that losing the lottery does push people out of the company and out of the US, but the magnitudes of the estimates are quite small. The fundamental problem is that it is difficult to distinguish whether my estimate of β_1 is small because the true population coefficient is small (either due to the multiple tries at the lottery or because firms find other ways to retain their employees) or due to measurement error.

Based on this 'trial run' and the empirical issues that have emerged, my advisors and I have decided that it would be better for this to not be my job market paper. Nonetheless we think it's an important and big enough question for my work to be a meaningful contribution to the field, so they've advised that I try to write up a draft quickly enough that I have time to pivot to an alternate JMP idea. I'm primarily looking for feedback on what the most compelling packaging/'story' for this paper is, that doesn't require too much additional data work beyond what I've already done.

Table 1: Application-level regressions of various labor market outcomes one year after the H-1B lottery on an indicator for winning the H-1B lottery. All regressions include a control for Advanced Degree Exemption (ADE) status, which increases lottery odds.

	work1yr		us1yr		educ1yr	
	(1)	(2)	(3)	(4)	(5)	(6)
win	0.0186***	0.0054**	0.0071***	0.0021*	-0.0082***	0.0006
	(0.0024)	(0.0025)	(0.0011)	(0.0012)	(0.0024)	(0.0026)
Firm x Year FEs	No	Yes	No	Yes	No	Yes
Obs.	77,606	77,606	77,606	77,606	77,606	77,606
DV mean	0.690	0.690	0.950	0.950	0.248	0.248

Table 2: Application-level regressions of various labor market outcomes two years after the H-1B lottery on an indicator for winning the H-1B lottery. All regressions include a control for Advanced Degree Exemption (ADE) status, which increases lottery odds.

	work2yr		us2yr		educ2yr	
	(1)	(2)	(3)	(4)	(5)	(6)
win	0.0052*	0.0010	0.0153***	0.0038**	-0.0049**	0.0029
	(0.0027)	(0.0027)	(0.0015)	(0.0016)	(0.0025)	(0.0027)
Firm x Year FEs	No	Yes	No	Yes	No	Yes
Obs.	77,606	77,606	77,606	77,606	77,606	77,606
DV mean	0.523	0.523	0.911	0.911	0.276	0.276